# KOSZUL LIFTING ALONG ESSENTIALLY FINITE PRESENTATION FLAT MAPS

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*Allowed standard prerequisites (editorial).*

- Stably free lifting. more-algebra-lemma-lift-complex-stably-frees (cross-chapter, named). Represents K by a bounded complex of free modules tensoring to the Koszul complex.
- Regular sequences. algebra-lemma-grothendieck-regular-sequence-general (cross-chapter, named). Lifts form a regular sequence with flat quotient.
- Resolution comparison. algebra-lemma-compare-resolutions (cross-chapter, named). Produces the dotted comparison arrow.

*Source reference.* Original reference 00LS: focused-seeds/geometry/S9-eqv-deform-koszul/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0470: focused-seeds/geometry/S9-eqv-deform-koszul/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0623: focused-seeds/geometry/S9-eqv-deform-koszul/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0BCB: focused-seeds/geometry/S9-eqv-deform-koszul/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 062F: focused-seeds/geometry/S9-eqv-deform-koszul/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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*Date*: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $(R, \mathfrak{m}, \kappa) \to (A, \mathfrak{n}, \lambda)$ be a flat local ring homorphism of local rings which is essentially of finite presentation. Let $\overline{f}_1, \ldots, \overline{f}_r \in \mathfrak{n}/\mathfrak{m}A \subset A/\mathfrak{m}A$ be a regular sequence. Let $K \in D(A)$. Assume*

(1) *$K$ is perfect,*
(2) *$K \otimes^{\mathbf{L}}_A A/\mathfrak{m}A$ is isomorphic in $D(A/\mathfrak{m}A)$ to the Koszul complex on $\overline{f}_1, \ldots, \overline{f}_r$.*

*Then $K$ is isomorphic in $D(A)$ to a Koszul complex on a regular sequence $f_1, \ldots, f_r \in A$ lifting the given elements $\overline{f}_1, \ldots, \overline{f}_r$. Moreover, $A/(f_1, \ldots, f_r)$ is flat over $R$.*

*Proof.* Let us use chain complexes in the proof of this lemma. The Koszul complex $K_\bullet(\overline{f}_1, \ldots, \overline{f}_r)$ is defined in More on Algebra, Definition 0623. By More on Algebra, Lemma 0BCB we can represent $K$ by a complex

$$K_\bullet : A \to A^{\oplus r} \to \ldots \to A^{\oplus r} \to A$$

whose tensor product with $A/\mathfrak{m}A$ is equal (!) to $K_\bullet(\overline{f}_1, \ldots, \overline{f}_r)$. Denote $f_1, \ldots, f_r \in A$ the components of the arrow $A^{\oplus r} \to A$. These $f_i$ are lifts of the $\overline{f}_i$. By Algebra, Lemma 0470 $f_1, \ldots, f_r$ form a regular sequence in $A$ and $A/(f_1, \ldots, f_r)$ is flat over $R$. Let $J = (f_1, \ldots, f_r) \subset A$. Consider the diagram

$$\begin{array}{ccc} K_\bullet & \xrightarrow{\varphi_\bullet} & K_\bullet(f_1, \ldots, f_r) \\ & \searrow \quad \swarrow & \\ & A/J & \end{array}$$

Since $f_1, \ldots, f_r$ is a regular sequence the south-west arrow is a quasi-isomorphism (see More on Algebra, Lemma 062F). Hence we can find the dotted arrow making the diagram commute for example by Algebra, Lemma 00LS. Reducing modulo $\mathfrak{m}$ we obtain a commutative diagram

$$\begin{array}{ccc} K_\bullet(\overline{f}_1, \ldots, \overline{f}_r) & \xrightarrow{\overline{\varphi}_\bullet} & K_\bullet(\overline{f}_1, \ldots, \overline{f}_r) \\ & \searrow \quad \swarrow & \\ & (A/\mathfrak{m}A)/(\overline{f}_1, \ldots, \overline{f}_r) & \end{array}$$

by our choice of $K_\bullet$. Thus $\overline{\varphi}$ is an isomorphism in the derived category $D(A/\mathfrak{m}A)$. It follows that $\overline{\varphi} \otimes^{\mathbf{L}}_{A/\mathfrak{m}A} \lambda$ is an isomorphism. Since $\overline{f}_i \in \mathfrak{n}/\mathfrak{m}A$ we see that

$$\mathrm{Tor}_i^{A/\mathfrak{m}A}(K_\bullet(\overline{f}_1, \ldots, \overline{f}_r), \lambda) = K_i(\overline{f}_1, \ldots, \overline{f}_r) \otimes_{A/\mathfrak{m}A} \lambda$$

Hence $\varphi_i \bmod \mathfrak{n}$ is invertible. Since $A$ is local this means that $\varphi_i$ is an isomorphism and the proof is complete. □